

The empirical recurrence for OEIS A125421, proved from an exact model

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Abstract

OEIS A125421 carries an empirical recurrence of order 3. It is true, and it is decidable rather than empirical. The entry's sequence is given exactly by the trace of a power of the adjacency matrix of the digit graph, from which a provably satisfies a MONIC linear recurrence of order $S = 9$, derived below and not assumed. The residual of the conjectured recurrence satisfies the same one, so whether it annihilates a is settled by a finite exact computation. The count is a walk count, but not of the usual shape: it is a TRACE and not a product with two boundary vectors, so the bound comes from the characteristic polynomial of the matrix itself.

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1 The conjecture

OEIS A125421 is “Number of base 9 circular n -digit numbers with adjacent digits differing by 7 or less.”. It has offset 0 and begins

1, 9, 79, 681, 5987, 52649, 463087, 4073225, ...

The entry states, as a conjecture contributed by its author and never marked settled:

[Empirical] $a(\text{base}, n) = a(\text{base}-1, n) + F(7)$ for $\text{base} \geq 7 \cdot \text{int}(n/2) + 1$ and $F(d)$ is the largest coefficient in $(1+x+\dots+x^{(2d)})^n$ $a(n) = 9 \cdot a(n-1) - a(n-2) - 7 \cdot a(n-3)$ for $n \geq 3$.

As of the “Last modified” line on the live entry (Aug 12 2026 19:06:38, revision 18) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a trace

Let G be the graph on the digits $0, 1, \dots, 8$ with an edge between u and v exactly when $|u-v| \leq 7$, loops included, and let M be its adjacency matrix, so $M_{uv} = 1$ when $|u-v| \leq 7$ and 0 otherwise. A circular n -digit string is a cyclic sequence $d_1, \dots, d_n$ in which every neighbouring pair, the pair (d_n, d_1) included, differs by at most 7 — that is, a CLOSED walk of length n in G together with the choice of where it starts. Closed walks of length n beginning at u number $(M^n)_{uu}$, so for $n \geq 1$

$$a(n) = \sum_{u=0}^8 (M^n)_{uu} = \text{tr } M^n.$$

The entry counts every string, leading zeros included: requiring a nonzero first digit gives 4, 11, 25 where A124698 gives 5, 13, 29. At $n = 0$ the entry writes $a(0) = 1$, the empty circular string, where the trace is 9; that is the entry's convention at one index and the two agree at every $n \geq 1$, on every published term.

3 The bound

tr M^n is the n -th power sum of the eigenvalues of M , so by Cayley–Hamilton the characteristic polynomial of M — monic of degree exactly $S = 9$, computed from the matrix and not estimated — annihilates a for $n \geq 1$. Nothing about this bound is derived from the data: it is the size of the digit alphabet.

4 The proof

Lemma 1. *Let $A(z) = z^S - \sum_{j < S} \alpha_j z^j$ be the monic annihilator derived above and $q(z) = z^3 - \sum_i c_i z^{3-i}$ the characteristic polynomial of the conjectured recurrence. The residual $r(n) = a(n) - \sum_i c_i a(n-i)$ is a linear combination of shifts of a , so A annihilates r as well. Since $A(0) \neq 0$ the recurrence it gives runs in both directions, and S consecutive zeros of r force r to vanish at every later index.*

Proof. A annihilates a , hence every shift of a , hence any linear combination of shifts; that is r . A monic recurrence with nonzero constant term determines each term from the S before it and each from the S after it, so a run of S zeros propagates. $\square$

Theorem 2.

$$a(n) = 9a(n-1) - a(n-2) - 7a(n-3)$$

for every $n > 3$.

Proof. The terms $a(0), \dots$ were computed exactly in integer arithmetic from the model, extended by A where the model itself was not evaluated, and $r(n)$ formed for each index. Those integers vanish from $n = 3+1$ onwards, and the run exceeds $S+3$, which by Lemma 1 settles every larger index at once. The bound is exact: at $n = 3$ the recurrence fails on the entry’s own published terms, so no larger range holds. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: it is built by reading the entry’s English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no model involved, and holds wherever the proved range and the data overlap.

Third, the derived annihilator A was itself tested on terms it had not been given: the model was evaluated beyond the S values that determine it and A reproduced them. A bound that is too small fails this test, which is why it is run.

References

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